Matemàtiques I

1r Batx

SOLUCIONS DE LA FITXA · UNITAT 2

Àlgebra

FentMates

Intenta resoldre cada exercici abans de mirar-ne la solució. Els resultats es donen de manera exacta (fraccions i arrels) i, quan cal, amb el valor aproximat. Pots comprovar les divisions, la factorització, Gauss, les inequacions i les regions amb les eines interactives dels apunts de la unitat.

1 Polinomis. Operacions

1.

a) $\boxed{-16x^2 - 17x + 10}$

b) $\boxed{15x^2 + 3x - 3}$

c) $\boxed{6x^4 - 10x^3 - 35x^2 + 23x}$

d) $\boxed{3x^5 - 2x^4 - 20x^2 - 16x + 32}$

e) $\boxed{x^3 - \frac{9x^2}{10} + \frac{4x}{15}}$

2. $(x+y+z)^2 = ((x+y)+z)^2 = (x+y)^2 + 2(x+y)z + z^2 = x^2 + 2xy + y^2 + 2xz + 2yz + z^2$. $\boxed{\text{Queda demostrada.}}$

3. $\boxed{16x^4 - 24x^3 + 25x^2 - 12x + 4}$ (amb la identitat de l'exercici anterior).

4.

a) $6x^4 + (3b-10)x^3 + (2a-5b)x^2 + abx$: $3b - 10 = -19$, $ab = -6 \Rightarrow \boxed{a = 2,\ b = -3}$

b) $2x^2 + 9x + 18 - 3a$: $\boxed{b = 9}$, $18 - 3a = 3 \Rightarrow \boxed{a = 5}$

5.

a) $c(x) = \boxed{x + \frac{2}{3}}$, $r(x) = \boxed{-x - \frac{19}{3}}$

b) $c(x) = \boxed{3x^2 - 3x - 5}$, $r(x) = \boxed{10x + 14}$

c) $c(x) = \boxed{\frac{1}{2}}$, $r(x) = \boxed{-3x^2 + \frac{9}{2}}$

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6.

a)

	3	2	1	−5
−2		−6	8	−18
	3	−4	9	−23

$c(x) = \boxed{3x^2 - 4x + 9}$, $r = \boxed{-23}$

b)

	3	0	−2	−1	4
−2		−6	12	−20	42
	3	−6	10	−21	46

$c(x) = \boxed{3x^3 - 6x^2 + 10x - 21}$, $r = \boxed{46}$

c)

	1	$-\frac{3}{2}$	-1	6
$-\frac{3}{2}$		$-\frac{3}{2}$	$\frac{27}{4}$	$-\frac{69}{8}$
	1	$-\frac{9}{2}$	$\frac{23}{4}$	$-\frac{21}{8}$

$c(x) = \boxed{\frac{x^2}{2} - \frac{9x}{4} + \frac{23}{8}}$, $r = \boxed{-\frac{21}{8}}$ (quocient de Ruffini dividit entre 2)

d)

	4	0	−3	0	5
5		20	100	485	2425
	4	20	97	485	2430

$c(x) = \boxed{-4x^3 - 20x^2 - 97x - 485}$, $r = \boxed{2430}$ (quocient de Ruffini dividit entre -1)

e)

	2	0	−12	−6	0	−117
3		6	18	18	36	108
	2	6	6	12	36	−9

$c(x) = \boxed{2x^4 + 6x^3 + 6x^2 + 12x + 36}$, $r = \boxed{-9}$

7.

a) $P(-3) = a - 3 = 0 \Rightarrow \boxed{a = 3}$

b) $P(-\frac{3}{2}) = \frac{3a}{2} - \frac{3}{2} = 0 \Rightarrow \boxed{a = 1}$

c) $P(\frac{4}{3}) = \frac{16a}{9} + \frac{160}{9} = 0 \Rightarrow \boxed{a = -10}$

d) $P(5) = b + 2930 = 0 \Rightarrow \boxed{b = -2930}$

(Amb el teorema del residu, $P(r)$ ha de ser 0, on r és l'arrel del divisor; és el mateix que obtenir residu 0 a Ruffini.)

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8. $M(2) = 16 - 20 + 8 - 4 = 0$: sí que és divisible.

Per Ruffini:

	2	−5	4	−4
2		4	−2	4
	2	−1	2	0

$\Rightarrow N(x) = 2x^2 - x + 2$

2 Factorització d'un polinomi. Teorema del residu

9.

a) $P(1) = 0$: sí **b)** $P(2) = 36$: no **c)** $P(-1) = -18$: no **d)** $P(-4) = 0$: sí **e)** $P(-3) = -44$: no

10. $P(-1) = 1 + 2 - 3 = 0$, $P(3) = 9 - 6 - 3 = 0$. No: un polinomi de grau 2 té com a màxim dues arrels, i ja les hem trobades: $P(x) = (x+1)(x-3)$.

11.

a) $x = -5, x = 2, x = 3$

b) $x = -\frac{1}{2}, x = 0, x = 2$ (triple)

c) $x = -1$ (triple), $x = -\frac{2}{3}, x = \frac{1}{2}$

d) $x = -2, x = 1, x = 4$ $((x-4)(x-1)(x+2))$

e) $x = -5, x = -2, x = -1$ $((x+1)(x+2)(x+5))$

f) $x = 0, x = \frac{1}{2}, x = 1, x = 2$ (doble) $(x(x-2)^2(x-1)(2x-1))$

12.

a) $(x-3)(x-1)(x+2)$ (arrels: $x = -2, x = 1, x = 3$)

b) $(x-1)^2(x+3)$ (arrels: $x = -3, x = 1$ (doble))

c) $(x-1)(x+1)^2(2x-3)$ (arrels: $x = -1$ (doble), $x = 1, x = \frac{3}{2}$)

d) $(x-2)(2x-1)(2x+3)$ (arrels: $x = -\frac{3}{2}, x = \frac{1}{2}, x = 2$)

e) $x^2(x-2)(x+2)$ (arrels: $x = -2, x = 0$ (doble), $x = 2$)

f) $x(x+1)^2(2x-1)(3x+1)$ (arrels: $x = -1$ (doble), $x = -\frac{1}{3}, x = 0, x = \frac{1}{2}$)

g) $(x-2)(x-1)(x+1)(x+2)(x^2+1)$ (arrels: $x = -2, x = -1, x = 1, x = 2$)

h) $t = x^2$: $\frac{t^2}{4} + \frac{3}{4}t + \frac{9}{16} = 0 \Rightarrow \Delta = \frac{9}{16} - \frac{9}{16} = 0$, $t = -\frac{3}{2} < 0$: no té arrels reals. $P(x) = \frac{1}{16}(2x^2+3)^2$

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13.

a) $P(2) = k + 8 = 0 \Rightarrow \boxed{k = -8}$

b) $P(-2) = -2k - 12 = 0 \Rightarrow \boxed{k = -6}$

c) $P(-4) = -63k + 252 = 0 \Rightarrow \boxed{k = 4}$

d) $P(3) = -6k + 36 = 0 \Rightarrow \boxed{k = 6}$

14. $P(-3) = -27 - 9(3k + 2) - 3 = -48 - 27k$.

a) $-48 - 27k = 0 \Rightarrow \boxed{k = -\frac{16}{9}}$ **b)** $-48 - 27k = 15 \Rightarrow \boxed{k = -\frac{7}{3}}$

15. $P(x) = ax^2+bx+c$: $c = -3$, $a+b-3 = -6$, $9a+3b-3 = 0 \Rightarrow a = 2, b = -5$: $\boxed{P(x) = 2x^2 - 5x - 3}$

16. $P(2) = 8m - 24 - 8 + 8 = 8m - 24 = 0 \Rightarrow \boxed{m = 3}$

17. $P(2) = 0$: $4a + 2b = -2$; $P(-3) = 0$: $9a - 3b = 33$. $\boxed{a = 2, b = -5}$ (i aleshores $P(x) = (x - 2)(x + 3)(x + 1)$).

18. Teorema del residu: $P(-2) = 32 - 72 + 8 + 12 + 3m = -20 + 3m = 12 \Rightarrow \boxed{m = \frac{32}{3}}$

3 Fraccions algebraiques

19.

a) $\frac{2(x-2)(x+2)}{x(x-1)(x+2)} = \boxed{\frac{2(x-2)}{x(x-1)}}$

b) $\frac{x(x-1)(x+1)}{x(x+1)(x+2)} = \boxed{\frac{x-1}{x+2}}$

c) $\frac{x^2(x-3)(x+1)}{x^2(x-3)(x+3)} = \boxed{\frac{x+1}{x+3}}$

d) $\frac{(x-1)^3}{x(x-1)^2} = \boxed{\frac{x-1}{x}}$

e) $\frac{(x-1)^2(x+1)(2x-3)}{(x-1)(x+2)(2x-3)} = \boxed{\frac{(x-1)(x+1)}{x+2}}$

f) $\frac{(x-1)^2(x+3)(2x-1)}{(x-1)(x+3)(2x-1)} = \boxed{x-1}$

g) $\frac{(x-3)(x-2)(x+3)}{(x-3)(x-2)^2} = \boxed{\frac{x+3}{x-2}}$

h) $\frac{(x-2)(x-1)^3}{(x-2)^2(x-1)^2} = \boxed{\frac{x-1}{x-2}}$

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20.

a) $\boxed{\dfrac{x+2}{(x-1)(x+1)}}$

b) $\boxed{\dfrac{3x+1}{x(x+1)}}$

c) $\boxed{\dfrac{2x-5}{4x(x-1)}}$

d) $\boxed{\dfrac{2(2x+1)}{x(x-1)(x+1)}}$

e) $\boxed{\dfrac{5x^2-3x+1}{(x-2)(x+2)}}$

f) $\boxed{\dfrac{2(x^2+x-1)}{(x-1)^2(x+1)}}$

21.

a) $\boxed{\dfrac{1}{4}}$

b) $\boxed{\dfrac{x+3}{(x-2)(x+1)}}$

c) $\boxed{\dfrac{(3x+2)(4x+1)}{(x-2)(x+2)}}$

d) $\boxed{-\dfrac{2}{3x^2}}$

e) $\boxed{-\dfrac{2x^2-3x-3}{x-1}}$

f) $\boxed{\dfrac{x^2-6x-1}{2(x+1)^2}}$

g) $\boxed{x}$

22.

a) $\boxed{\dfrac{3x^4+3x^3+3x^2+4x+2}{x(x+2)(x^2+1)}}$

b) $\boxed{\dfrac{3x(x+1)}{x^2+1}}$

c) $\boxed{-\dfrac{a+b}{b^2}}$

d) $\boxed{-\dfrac{a(b^2-b-1)}{b^2}}$

e) $\boxed{\dfrac{2(5x+6)}{(x-2)(x+2)}}$

f) $\boxed{\dfrac{(x-2)(x+5)}{x+1}}$

g) $\boxed{xy(x-y)}$

h) $\boxed{\dfrac{1}{a+x}}$

i) $\boxed{\dfrac{3x+2}{2x+1}}$

j) $\boxed{\dfrac{3x^2+3x+2}{x^2+1}}$

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4 Equacions polinòmiques

23.

a) $\boxed{x = \frac{9}{32}}$

b) $\boxed{x = -5,\ x = 0}$

c) $\boxed{x = -\frac{\sqrt{30}}{6},\ x = \frac{\sqrt{30}}{6}}$

d) $\boxed{x = -2,\ x = \frac{11}{4}}$

e) $(x-1)(x+2)(2x+3)(3x-1) = 0$: $\boxed{x = -2,\ x = -\frac{3}{2},\ x = \frac{1}{3},\ x = 1}$

f) $(x-2)(2x+3)(x^2+3) = 0$: $\boxed{x = -\frac{3}{2},\ x = 2}$

g) $x(x+1)(2x-1)(3x+1) = 0$: $\boxed{x = -1,\ x = -\frac{1}{3},\ x = 0,\ x = \frac{1}{2}}$

h) $(x-2)(x-1)(x+3) = 0$: $\boxed{x = -3,\ x = 1,\ x = 2}$

i) $(x-3)^2(x+3) = 0$: $\boxed{x = -3,\ x = 3}$

j) $(x-3)(x+2)(x^2+1) = 0$: $\boxed{x = -2,\ x = 3}$

k) $(x-1)(x+1)(x^2+1) = 0$: $\boxed{x = -1,\ x = 1}$

l) $x(x-3)(x+1)(x+3) = 0$: $\boxed{x = -3,\ x = -1,\ x = 0,\ x = 3}$

24.

a) $t = x^2$: $t^2 - 17t + 16 = 0 \Rightarrow t = 16,\ t = 1$: $\boxed{x = \pm 4,\ x = \pm 1}$

b) $t = x^3$: $t^2 - 9t + 8 = 0 \Rightarrow t = 8,\ t = 1$: $\boxed{x = 2,\ x = 1}$

c) $t^2 - 8t - 9 = 0 \Rightarrow t = 9$ ($t = -1$ no dona solucions): $\boxed{x = \pm 3}$

d) $t = x^3$: $t^2 + 2t - 3 = 0 \Rightarrow t = 1,\ t = -3$: $\boxed{x = 1,\ x = -\sqrt[3]{3}}$

e) Desenvolupant: $2x^4 + 12x^2 - 14 = 0 \Rightarrow t^2 + 6t - 7 = 0 \Rightarrow t = 1$ ($t = -7$ no dona solucions): $\boxed{x = \pm 1}$

f) Multipliquem per x^2: $x^4 - 10x^2 + 9 = 0 \Rightarrow t = 9,\ t = 1$: $\boxed{x = \pm 3,\ x = \pm 1}$

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25.

a) $\boxed{x=-2,\ x=-\sqrt{3},\ x=\sqrt{3},\ x=2}$

b) $\boxed{x=-3,\ x=-\frac{2}{3},\ x=0,\ x=3,\ x=4}$

c) $\boxed{x=-3,\ x=2}$

d) $\boxed{x=-1,\ x=2,\ x=\frac{7}{2}}$

5 Equacions racionals

26.

a) $8x^2+8x+8=7x^3 \Rightarrow (x-2)(7x^2+6x+4)=0$ ($\Delta<0$): $\boxed{x=2}$

b) Multipliquem per $3(x-1)$: $2x^2+4x+9=11 \Rightarrow x^2+2x-1=0$: $\boxed{x=-1\pm\sqrt{2}}$

c) $2x(x+2)+3x(x-2)=6x^2 \Rightarrow x^2+2x=0 \Rightarrow x=0,\ x=-2$; $x=-2$ anul·la un denominador: $\boxed{x=0}$

d) m.c.m. $=(x-1)(x-2)(x-4)$: $3(x-1)+(x-5)(x-2)=(x-1)(x-2) \Rightarrow x^2-4x+7=x^2-3x+2$: $\boxed{x=5}$

e) Multipliquem per $\sqrt{2}\,x$: $x^2+2=2x^2 \Rightarrow \boxed{x=\pm\sqrt{2}}$

f) $-\dfrac{x}{x-2}=-x \Rightarrow x=x(x-2) \Rightarrow x^2-3x=0$: $\boxed{x=0,\ x=3}$

g) Simplificant: $\dfrac{2x(x+1)}{(x-1)(x+3)}=2 \Rightarrow x^2+x=x^2+2x-3$: $\boxed{x=3}$

h) El primer membre és $\dfrac{x+1}{x}$: és una $\boxed{\text{identitat}}$, es compleix per a tot $x\neq 0,\ x\neq -1$.

6 Equacions amb radicals

27.

a) $2x-3=(1-x)^2 \Rightarrow (x-2)^2=0$; $x=2$: $-1+1=0\neq 2$: $\boxed{\text{no té solució}}$

b) $2x^2+2x-3=(x+1)^2 \Rightarrow x^2=4$; $x=2$ no val ($-1-2<0$): $\boxed{x=-2}$

c) $9(7x+1)=(5x-7)^2 \Rightarrow 25x^2-133x+40=0 \Rightarrow x=5,\ x=\frac{8}{25}$; $x=\frac{8}{25}$ no val: $\boxed{x=5}$

d) $2x-1=12$: $\boxed{x=\dfrac{13}{2}}$

e) $2-5x=3x^2 \Rightarrow 3x^2+5x-2=0 \Rightarrow x=-2,\ x=\frac{1}{3}$; $x=\frac{1}{3}$ no val: $\boxed{x=-2}$

f) $2+\sqrt{x-4}=12-x \Rightarrow x-4=(10-x)^2 \Rightarrow x^2-21x+104=0 \Rightarrow x=8,\ x=13$; $x=13$ no val: $\boxed{x=8}$

g) $x=4-4\sqrt{x-4}+x-4 \Rightarrow \sqrt{x-4}=0$: $\boxed{x=4}$

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h) $x-46=-14\sqrt{x+5} \Rightarrow x^2-288x+1136=0 \Rightarrow x=4, x=284$; $x=284$ no val: $\boxed{x=4}$

i) $10\sqrt{x-1}=20 \Rightarrow \sqrt{x-1}=2$: $\boxed{x=5}$

j) $2\sqrt{2x^2-14x}=8-2x \Rightarrow x^2-6x-16=0 \Rightarrow x=8, x=-2$; cap no val: $\boxed{\text{no té solució}}$

k) $\sqrt{1-4x}=3\sqrt{3+x} \Rightarrow 1-4x=27+9x$: $\boxed{x=-2}$

28.

a) $x^2-1=(x-1)^3 \Rightarrow x(x-1)(x-3)=0$; totes valen ($\sqrt[3]{-1}=-1$): $\boxed{x=0, x=1, x=3}$

b) $\sqrt{3x+7}=x+1 \Rightarrow x^2-x-6=0 \Rightarrow x=3, x=-2$; $x=-2$ no val: $\boxed{x=3}$

c) $4x-1=(x-4)^3 \Rightarrow (x-7)(x^2-5x+9)=0$: $\boxed{x=7}$

d) $(4-2x)^2=8x^2-16x \Rightarrow 4x^2=16$; totes dues valen: $\boxed{x=2, x=-2}$

e) $(2x+2)^2=6x+10 \Rightarrow 2x^2+x-3=0 \Rightarrow x=1, x=-\frac{3}{2}$; $x=-\frac{3}{2}$ no val: $\boxed{x=1}$

29. $\sqrt{x+1}=2\sqrt{x-11} \Rightarrow x+1=4x-44$: $\boxed{x=15}$ (comprovació: $\sqrt{16}=4=2\sqrt{4}$).

7 Equacions logarítmiques

30.

a) $2(x+3)=x+5$: $\boxed{x=-1}$

b) $15-2x=x^2 \Rightarrow x=3, x=-5$; $x=-5$ no val: $\boxed{x=3}$

c) $\dfrac{2x+3}{x-1}=36 \Rightarrow 34x=39$: $\boxed{x=\dfrac{39}{34}}$

d) $x^2-3x+2=x^2 \Rightarrow \boxed{x=\dfrac{2}{3}}$ (val: $x>0$ i $x^2-3x+2=\frac{4}{9}>0$)

e) $\dfrac{2x-2}{x}=\dfrac{(x-1)^2}{x} \Rightarrow (x-1)(x-3)=0$; $x=1$ no val: $\boxed{x=3}$

f) $\log x^2=\log\dfrac{x}{20} \Rightarrow x^2=\dfrac{x}{20}$; $x=0$ no val: $\boxed{x=\dfrac{1}{20}}$

g) $2x(4-5x)=3(2x-x^2) \Rightarrow 7x^2-2x=0$; $x=0$ no val: $\boxed{x=\dfrac{2}{7}}$

h) $\dfrac{5(x+1)(x+2)}{x+2}=4 \Rightarrow 5x+5=4$: $\boxed{x=-\dfrac{1}{5}}$

i) $4-x=(x+2)^2 \Rightarrow x^2+5x=0$; $x=-5$ no val: $\boxed{x=0}$

j) Canvi de base: $\dfrac{2\log 3}{\log(x+3)}=\dfrac{\log 3}{\log 5} \Rightarrow \log(x+3)=2\log 5 \Rightarrow x+3=25$: $\boxed{x=22}$

k) En base 3: $\log_3 32-\log_3(6-x)=2\log_3 x \Rightarrow x^2(6-x)=32 \Rightarrow (x+2)(x-4)^2=0$: $\boxed{x=4}$

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31. $\log x + \log 2 = 1 \Rightarrow \log(2x) = \log 10$: $\boxed{x = 5}$

32. $2\log x = \log 4 + \log 9 \Rightarrow x^2 = 36$ ($x > 0$): $\boxed{x = 6}$

8 Equacions exponencials

33.

a) $2^{2x^2+2} = 2^{5x+5} \Rightarrow 2x^2 - 5x - 3 = 0$: $\boxed{x = 3,\ x = -\frac{1}{2}}$

b) $262144 = 4^9 \Rightarrow (x-2)^2 = 9$: $\boxed{x = 5,\ x = -1}$

c) $x + 4 = \frac{\log 5}{\log 3}$: $\boxed{x = \frac{\log 5}{\log 3} - 4 \approx -2{,}535}$

d) $2^{-x} = 2^{2x(x-1)} \Rightarrow 2x^2 - x = 0$: $\boxed{x = 0,\ x = \frac{1}{2}}$

e) $13^{x^2+2x} = 13^{-1} \Rightarrow (x+1)^2 = 0$: $\boxed{x = -1}$

f) $2x^2 - 4x - 2 = 4 \Rightarrow x^2 - 2x - 3 = 0$: $\boxed{x = 3,\ x = -1}$

g) $6^{4x-3} = 12 \Rightarrow 4x - 3 = \frac{\log 12}{\log 6}$: $\boxed{x = \frac{3 + \log_6 12}{4} \approx 1{,}097}$

h) $\left(\frac{3}{2}\right)^x = \frac{3}{2}$: $\boxed{x = 1}$

i) $3^{3/x} = 3^{x+2} \Rightarrow x^2 + 2x - 3 = 0$: $\boxed{x = 1,\ x = -3}$

j) $2^{x+4} = 2^{3x} \Rightarrow x + 4 = 3x$: $\boxed{x = 2}$

k) $x^2 - 5x + 6 = 0$: $\boxed{x = 2,\ x = 3}$

l) $7^{\frac{2(x-5)}{3}} = 7^{-\frac{x-1}{2}} \Rightarrow 4(x-5) = -3(x-1)$: $\boxed{x = \frac{23}{7}}$

34.

a) $2^x\left(2 + 1 + \frac{1}{2}\right) = 28 \Rightarrow 2^x = 8$: $\boxed{x = 3}$

b) $5^x\left(125 - \frac{1}{5}\right) = 3120 \Rightarrow 5^x = 25$: $\boxed{x = 2}$

c) $4^x \cdot \frac{21}{16} = 336 \Rightarrow 4^x = 256$: $\boxed{x = 4}$

d) $3^x\left(9 + 3 + 1 + \frac{1}{3}\right) = 120 \Rightarrow 3^x = 9$: $\boxed{x = 2}$

e) $2^x \cdot \frac{15}{16} = 960 \Rightarrow 2^x = 1024$: $\boxed{x = 10}$

f) $t = 6^x$: $t^2 + 8t - 9 = 0 \Rightarrow t = 1$ ($t = -9$ no val): $\boxed{x = 0}$

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35.

a) $t = 3^x$: $3t^2 - 10t + 3 = 0 \Rightarrow t = 3, t = \frac{1}{3}$: $\boxed{x = 1, x = -1}$

b) $t = 10^{x+2}$: $2t^2 + 3t - 5 = 0 \Rightarrow t = 1$ ($t = -\frac{5}{2}$ no val) $\Rightarrow x + 2 = 0$: $\boxed{x = -2}$

c) $t = 3^x$: $t^2 + 5t - 24 = 0 \Rightarrow t = 3$ ($t = -8$ no val): $\boxed{x = 1}$

d) $t = 3^x$: $t^2 + \frac{t^2}{3} + \frac{t}{3} = 111 \Rightarrow 4t^2 + t - 333 = 0 \Rightarrow t = 9$: $\boxed{x = 2}$

e) $t = 3^x$: $9t + \frac{t^2}{9} = 90 \Rightarrow t^2 + 81t - 810 = 0 \Rightarrow t = 9$ ($t = -90$ no val): $\boxed{x = 2}$

f) $t = 2^x$: $\frac{t^2}{16} - \frac{5t}{8} + 1 = 0 \Rightarrow t^2 - 10t + 16 = 0 \Rightarrow t = 8, t = 2$: $\boxed{x = 3, x = 1}$

g) $t = 3^x$: $81t^2 + 27t - 810 = 0 \Rightarrow 3t^2 + t - 30 = 0 \Rightarrow t = 3$ ($t = -\frac{10}{3}$ no val): $\boxed{x = 1}$

h) $t = 2^x$: $4t^2 + 8t - 320 = 0 \Rightarrow t^2 + 2t - 80 = 0 \Rightarrow t = 8$ ($t = -10$ no val): $\boxed{x = 3}$

9 Problemes d'equacions

36. $5x + 2 \cdot 50 + 0 \cdot 50 = 4(x + 100) \Rightarrow \boxed{x = 300 \text{ L}}$

37. $30 \cdot 5 + 10x = 7(30 + x) \Rightarrow 3x = 60$: $\boxed{20 \text{ L}}$

38. En 1 kg, x kg de te blanc: $18x + 14(1 - x) = 16{,}4 \Rightarrow x = 0{,}6$: $\boxed{60\,\% \text{ blanc i } 40\,\% \text{ verd}}$

39. El 10 % són 15: $\boxed{150 \text{ alumnes}}$

40. A la venda queda el $75\,\%$ del $20\,\%$, és a dir, el $15\,\%$: $0{,}15x = 18\,000 \Rightarrow \boxed{120\,000 \text{ entrades}}$

41. Àfrica és el 15 %: $0{,}15x = 3 \Rightarrow \boxed{20 \text{ milions d'euros}}$

42. $5x - 2(20 - x) = 58 \Rightarrow 7x = 98$: $\boxed{14 \text{ encerts i } 6 \text{ errades}}$

43. $(x - 20)\left(\frac{125}{x} + 0{,}4\right) = 147 \Rightarrow x^2 - 75x - 6250 = 0 \Rightarrow x = 125$ ($x = -50$ no té sentit): $\boxed{125 \text{ kg}}$

44. $x^2 + (x + 1)^2 = 25 \Rightarrow x^2 + x - 12 = 0 \Rightarrow x = 3$: $\boxed{3 \text{ cm i } 4 \text{ cm}}$

45. $x^2 + (x + 2)^2 = (x + 4)^2 \Rightarrow x^2 - 4x - 12 = 0 \Rightarrow x = 6$: $\boxed{6,\ 8 \text{ i } 10 \text{ cm}}$

46. $x + y = 70$, $x^2 + y^2 = 2500 \Rightarrow x^2 - 70x + 1200 = 0$: $\boxed{40 \text{ m} \times 30 \text{ m}}$

47. $\frac{1}{5} + \frac{1}{20} = \frac{1}{4}$: $\boxed{T = 4 \text{ h}}$

48. $\frac{1}{2} + \frac{1}{3} = \frac{5}{6} \Rightarrow T = \frac{6}{5}$ h: $\boxed{1 \text{ h } 12 \text{ min}}$

49. $\frac{1}{t} + \frac{1}{2t} = \frac{1}{10} \Rightarrow t = 15$: $\boxed{15 \text{ h i } 30 \text{ h}}$

50. $\frac{1}{10} + \frac{1}{20} + \frac{1}{30} + \frac{1}{40} + \frac{1}{50} = \frac{137}{600} \Rightarrow T = \frac{600}{137} \approx \boxed{4{,}38 \text{ h} \approx 4 \text{ h } 23 \text{ min}}$

51. $\frac{600}{x - 10} = \frac{600}{x} + 5 \Rightarrow x^2 - 10x - 1200 = 0 \Rightarrow x = 40$: $\boxed{40 \text{ persones}}$

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52.

a) $\boxed{y(0) = 300 \text{ malalts}}$

b) $200e^{0,2t} = 800 \Rightarrow t = 5\ln 4 \approx 6{,}93$: $\boxed{\text{el 7 de gener (cap al final del dia)}}$

53.

a) $e^{-0,1t} = \frac{1}{2} \Rightarrow t = 10\ln 2 \approx \boxed{6{,}93 \text{ dies}}$

b) $27e^{-1} \approx \boxed{9{,}93 \text{ mg}}$

54. $2^t = \dfrac{100}{9} \Rightarrow t = \dfrac{\log(100/9)}{\log 2} \approx \boxed{3{,}47 \text{ h} \approx 3 \text{ h } 28 \text{ min}}$

55.

a) $1 + Ka = 4,\ 1 + Ka^2 = 2{,}5 \Rightarrow Ka = 3,\ Ka^2 = 1{,}5$: $\boxed{a = \dfrac{1}{2},\ K = 6}$

b) $1 + 6 \cdot 0{,}5^{x/10} = \frac{4}{3} \Rightarrow 0{,}5^{x/10} = \frac{1}{18} \Rightarrow x = 10\log_2 18 \approx \boxed{41{,}7 \text{ anys}}$

10 Sistemes d'equacions lineals. Mètode de Gauss

56.

a) $2x + 10 = \frac{x+1}{2} + 3$: $\boxed{x = -\dfrac{13}{3},\ y = \dfrac{4}{3}}$

b) $-5x + 8y = -34,\ 3x - 2y = 12$: $\boxed{x = 2,\ y = -3}$

57.

a) SCD: $\boxed{(x,y,z) = (2, 1, -1)}$

b) SI: incompatible (apareix una equació $0 = k \neq 0$)

c) SCI: $\boxed{(x,y,z) = (-\lambda + 2, \lambda + 1, \lambda),\ \lambda \in \mathbb{R}}$

d) SCI: $\boxed{(x,y,z) = (\dfrac{3\lambda}{5} + \dfrac{4}{5}, \dfrac{7\lambda}{5} - \dfrac{4}{5}, \lambda),\ \lambda \in \mathbb{R}}$

e) SI: incompatible (apareix una equació $0 = k \neq 0$)

f) SCI: $\boxed{(x,y,z) = (-\dfrac{7\lambda}{4}, \dfrac{5\lambda}{4}, \lambda),\ \lambda \in \mathbb{R}}$

g) SCD: $\boxed{(x,y,z) = (1, 0, 1)}$

h) SCD: $\boxed{(x,y,z) = (-2, 3, 1)}$

i) SI: incompatible (apareix una equació $0 = k \neq 0$)

j) SCD: $\boxed{(x,y,z,w) = (4, 2, -1, 0)}$

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11 Sistemes d'equacions no lineals

58.

a) $x = \frac{-4-3y}{2}$ i substituïm: $\boxed{(-\frac{15}{7}, \frac{2}{21}), (1, -2)}$

b) $\frac{y+2x}{xy} = 4 \Rightarrow y + 2x = \frac{18}{5}$: $\boxed{(\frac{3}{10}, 3), (\frac{3}{2}, \frac{3}{5})}$

c) Reducció: $7x^2 = 28$: $\boxed{(-2, -1), (-2, 1), (2, -1), (2, 1)}$

d) Restant: $3y^2 - y - 14 = 0$: $\boxed{(\frac{17}{6}, \frac{7}{3}), (5, -2)}$

e) $x = \frac{-8-5y}{2}$ i substituïm: $\boxed{(-2, -\frac{4}{5}), (1, -2)}$

f) $-x^2 = 1$: $\boxed{\text{no té solució real}}$

g) $y = 2x - 3 \Rightarrow x^2 - 4x + 4 = 0$: $\boxed{(2, 1)}$

h) $y + 1 = 12 - x \Rightarrow x^2 - 7x + 12 = 0$: $\boxed{(3, 8), (4, 7)}$

i) $y = -\frac{3}{x} \Rightarrow x^4 - 19x^2 + 18 = 0$: $\boxed{(-3\sqrt{2}, \frac{\sqrt{2}}{2}), (-1, 3), (1, -3), (3\sqrt{2}, -\frac{\sqrt{2}}{2})}$

j) Reducció: $17y^2 = 0$: $\boxed{(-4, 0), (4, 0)}$

k) $(\sqrt{x} - \sqrt{y})(\sqrt{x} + \sqrt{y}) = 105 \Rightarrow \sqrt{x} - \sqrt{y} = 7$: $\boxed{(121, 16)}$

l) $(x - y)^2 = 16 \Rightarrow x - y = \pm 4$: $\boxed{(-11, -7), (11, 7)}$

59.

a) $u + v = 9, 4u + 5v = 41 \Rightarrow u = 4, v = 5$: $\boxed{x = 2, y = 1}$

b) $u + v = 5, 2u - v = 7 \Rightarrow u = 4, v = 1$: $\boxed{x = 2, y = 0}$

c) $2^x = 128 \cdot 2^y \Rightarrow 127 \cdot 2^y = 1016 \Rightarrow 2^y = 8$: $\boxed{x = 10, y = 3}$

d) $u + v = 16, \frac{u}{3} - 49v = -340 \Rightarrow u = 9, v = 7$: $\boxed{x = 2, y = 1}$

e) $y = 5 - x$ i $t = 2^x$: $\frac{1}{t} + \frac{3t}{32} = \frac{5}{8} \Rightarrow 3t^2 - 20t + 32 = 0 \Rightarrow t = 4, t = \frac{8}{3}$:

$\boxed{(2, 3)}$ i $\boxed{\left(\log_2 \frac{8}{3}, 5 - \log_2 \frac{8}{3}\right) \approx (1{,}415;\ 3{,}585)}$

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60.

a) $2u-3v=7,\ u+v=1 \Rightarrow u=2,\ v=-1$: $\boxed{x=100,\ y=\dfrac{1}{10}}$

b) $xy=100,\ x=21+y \Rightarrow y^2+21y-100=0 \Rightarrow y=4$ ($y=-25$ no val): $\boxed{x=25,\ y=4}$

c) $u+3v=5,\ u-v=3 \Rightarrow u=\frac{7}{2},\ v=\frac{1}{2}$: $\boxed{x=10^{7/2}=1000\sqrt{10},\ y=\sqrt{10}}$

d) $xy=3^7=2187$ i $x=27+2y \Rightarrow 2y^2+27y-2187=0 \Rightarrow y=27$: $\boxed{x=81,\ y=27}$

e) $x=10y \Rightarrow 32y=64$: $\boxed{x=20,\ y=2}$

f) $2u-2v=6,\ 2u+2v=2 \Rightarrow u=2,\ v=-1$: $\boxed{x=100,\ y=\dfrac{1}{10}}$

12 Inequacions i sistemes amb una incògnita

61.

a) $-8+4-2=-6\leq 6$: $\boxed{\text{sí}}$

b) $\frac{1}{e}-1\approx -0{,}63\not\geq 0$: $\boxed{\text{no}}$

c) $2^{-0{,}5}-2{,}5\approx -1{,}79<3^{-0{,}5}\approx 0{,}58$: $\boxed{\text{sí}}$

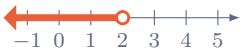
62.

a) $4x-12-x+2\leq 4x \Rightarrow -x\leq 10$: $\boxed{x\in[-10,\ +\infty)}$



b) $9x-18>9x+1 \Rightarrow -18>1$, fals: $\boxed{\text{no té solució}}$

c) $6x+8<\frac{x+38}{2} \Rightarrow 11x<22$: $\boxed{x\in(-\infty,2)}$

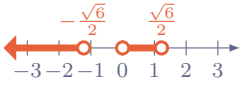


d) $2x^2-8x\leq 0 \Rightarrow 2x(x-4)\leq 0$: $\boxed{x\in[0,4]}$



63.

a) $-x(2x^2-3)>0$: $\boxed{x\in(-\infty,-\dfrac{\sqrt{6}}{2})\cup(0,\dfrac{\sqrt{6}}{2})}$



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b) $(x+3)(3x-8) \geq 0$: $\boxed{x \in (-\infty, -3] \cup [\frac{8}{3}, +\infty)}$



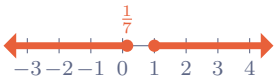
c) $x(x-2)(x+2) > 0$: $\boxed{x \in (-2, 0) \cup (2, +\infty)}$



d) $(x-1)(x+1)^2 \leq 0$: $\boxed{x \in (-\infty, 1]}$



e) $(x-1)(7x-1) \geq 0$: $\boxed{x \in (-\infty, \frac{1}{7}] \cup [1, +\infty)}$



f) $-(x-1)(x+1)(x+2) \leq 0$: $\boxed{x \in [-2, -1] \cup [1, +\infty)}$



g) $(x-1)(x+1)(x^2+1) > 0$: $\boxed{x \in (-\infty, -1) \cup (1, +\infty)}$

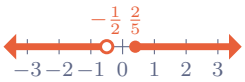


h) $(x+1)(x+2) < 0$: $\boxed{x \in (-2, -1)}$



64.

a) $\dfrac{5x-2}{2x+1} \geq 0$: $\boxed{x \in (-\infty, -\frac{1}{2}) \cup [\frac{2}{5}, +\infty)}$



b) $\dfrac{(x-2)(x+1)}{(x-1)(x+2)} \geq 0$: $\boxed{x \in (-\infty, -2) \cup [-1, 1) \cup [2, +\infty)}$



c) $\dfrac{4x-5}{(x+1)(4x-5)} < 0$: $\boxed{x \in (-\infty, -1)}$



d) $\dfrac{x^3}{x-9} \geq 0$: $\boxed{x \in (-\infty, 0] \cup (9, +\infty)}$

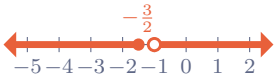


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e) $\dfrac{(x-1)(x+5)}{x(x-1)(x+1)} \geq 0$: $\boxed{x \in [-5,-1) \cup (0,1) \cup (1,+\infty)}$

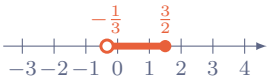


f) $\dfrac{2x+3}{(x+1)(x^2-x+1)} \geq 0$: $\boxed{x \in (-\infty,-\frac{3}{2}] \cup (-1,+\infty)}$



65.

a) $x \leq \frac{3}{2}$ i $x > -\frac{1}{3}$: $\boxed{x \in (-\frac{1}{3},\frac{3}{2}]}$



b) $x \leq 0$, $x \leq -1$ i $x \in (-\infty,-4) \cup (2,+\infty)$: $\boxed{x \in (-\infty,-4)}$



c) $x < 1$ i $x \geq -1$: $\boxed{x \in [-1,1)}$



d) $x > 2$ i $x \leq 1$: $\boxed{\text{no té solució}}$

e) $(-\infty,-3] \cup [1,+\infty)$ i $x > 0$: $\boxed{x \in [1,+\infty)}$



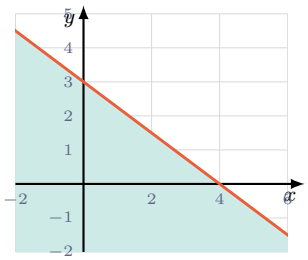
f) $(-\infty,-2) \cup (1,+\infty)$ i $(-\infty,-1] \cup [1,+\infty)$: $\boxed{x \in (-\infty,-2) \cup (1,+\infty)}$



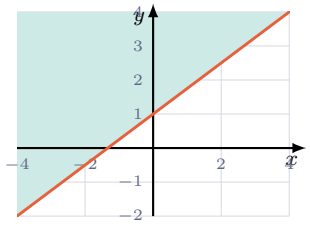
13 Inequacions i sistemes amb dues incògnites

66.

a) Recta contínua per $(4,0)$ i $(0,3)$; O: $0 \leq 12$ ✓: semiplà de O.

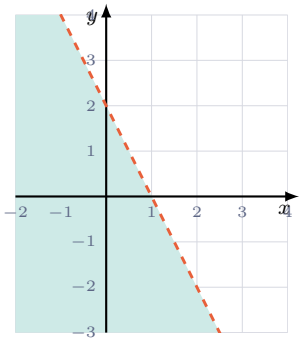


b) Recta contínua per $(0,1)$ i $(-\frac{4}{3},0)$; O: $0 \geq 4$ fals: semiplà que $\boxed{\text{no}}$ conté O.

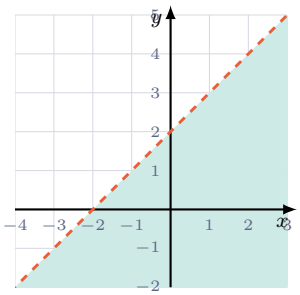


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c) Recta discontínua per $(1,0)$ i $(0,2)$; O: $0 < 2$ ✓: semiplà de O.

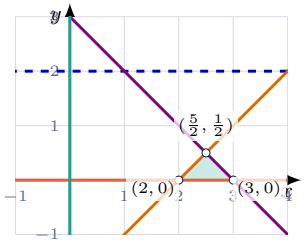


d) $x-y > -2$: recta discontínua per $(-2,0)$ i $(0,2)$; O: $0 > -2$ ✓: semiplà de O.

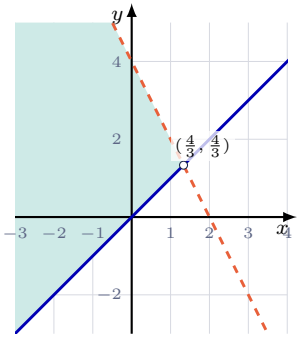


67.

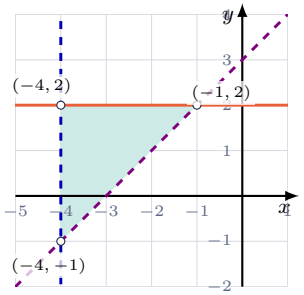
a) Triangle de vèrtexs $(2,0)$, $(3,0)$ i $\left(\frac{5}{2},\frac{1}{2}\right)$ (la condició $y < 2$ no hi influeix).



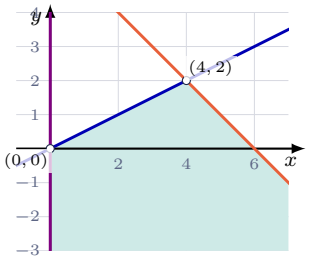
b) Regió no fitada amb un sol vèrtex, $\left(\frac{4}{3},\frac{4}{3}\right)$ (la vora de $y < -2x + 4$ no s'inclou).



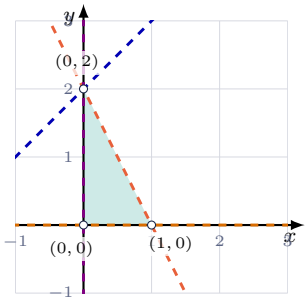
c) Triangle de vèrtexs $(-4,-1)$, $(-4,2)$ i $(-1,2)$; només el costat $y = 2$ és de la regió.



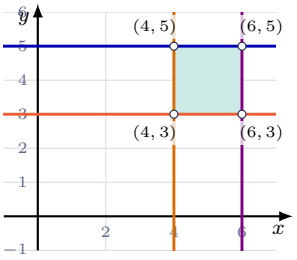
d) Regió no fitada (cap avall) amb vèrtexs $(0,0)$ i $(4,2)$.



e) Triangle de vèrtexs $(0,0)$, $(1,0)$ i $(0,2)$, sense les vores ($x > y - 2$ no hi influeix).



f) Rectangle de vèrtexs $(4,3)$, $(6,3)$, $(6,5)$ i $(4,5)$.



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14 Problemes de sistemes i inequacions

68. $x+y=250$, $4x+2y=800$: 150 conills i 100 gallines

69. $x^2h=2$, $2x^2+4xh=10 \Rightarrow 2x^2+\frac{8}{x}=10 \Rightarrow x^3-5x+4=0 \Rightarrow (x-1)(x^2+x-4)=0$. L'única arrel entera és $x=1$: 1 dm × 1 dm × 2 dm

70. $x+y+z=21$, $50x+100y+200z=1800$, $x=5z$: 10 de 50 £, 9 de 100 £, 2 de 200 £

71. $L+C+P=60$, $L=3(C+P)$, $2C=3P$: Lluís 45 £, Carme 9 £, Pere 6 £

72. $b+t+c=50$, $b+t=4c$, $2b+3t+4c=140$: 20 bicicletes, 20 tricicles, 10 cotxes

73. $10a+15b+30c=775$, $b=c+5$, $c=0{,}2(a+b+c)$: 25 A, 15 B, 10 C

74. $x+y+z=8$, $10x+20y+50z=290$, $x=2y$: 2 de 10 £, 1 de 20 £, 5 de 50 £

75. $x+y+z=800$, $10x+20y+50z=21000$, $x=y+z$: 400 de 10 £, 100 de 20 £, 300 de 50 £

76. $n+e+j=1405$, $6n+4e+3j=7920$, $4e+3j=0{,}1\cdot 6n$:
1200 normals, 105 de l'espectador, 100 de jubilats

77. $a+b+c=200$, $b=a+c$, $200a+500b+250c=73500$: 30 A, 100 B, 70 C

78. $m+l+t=9$, $100m+120l+150t=1160$, $t=l+1$: 2 memòries, 3 lectors, 4 tauletes

79. $0{,}9(r+q+c)=3{,}96$, $q=\frac{r}{2}$, $c=q+0{,}2r$: retolador 2 £, quadern 1 £, carpeta 1,40 £

80. $a+b+d=7000$, $0{,}05a+0{,}03b+0{,}01d=202$, $a+b=d+2600$: A 1800 £, B 3000 £, dipòsit 2200 £

81. $l+p+r=60$, $p=2(l+r)$, $r=\frac{l}{2}+2$: Londres 12, París 40, Roma 8

82. $200b+100p+300q=830$, $1{,}25(200b+100p)+1{,}4\cdot 300q=1087$, $b=q+0{,}5$:
bresquilles 1,60 £/kg, pomes 1,80 £/kg, peres 1,10 £/kg

83.

a) $l=d+s$, $4d+8l+10s=1116$, $8l-10s=156$: 30 desdejunis, 72 dinars, 42 sopars

b) $30\cdot 2{,}5+72\cdot 4+42\cdot 5=$ 573 £

84. $p_1+p_2+p_3=7{,}5$, $p_3=p_2+1$, $p_1=5(p_3-p_1)$: 2,5; 2; 3 punts

85. $a+b+c=42000$, $0{,}05a+0{,}07b+0{,}09c=2600$, $0{,}05a=0{,}07b+0{,}09c-200$: 24 000 £, 11 000 £, 7000 £

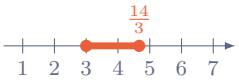
86. $10+0{,}05x<0{,}12x \Rightarrow x>\frac{1000}{7}\approx 142{,}9$: a partir de 143 minuts

87. $1{,}2+0{,}7x\geq 5 \Rightarrow x\geq \frac{38}{7}\approx 5{,}43$: $x\in[\frac{38}{7},10]$



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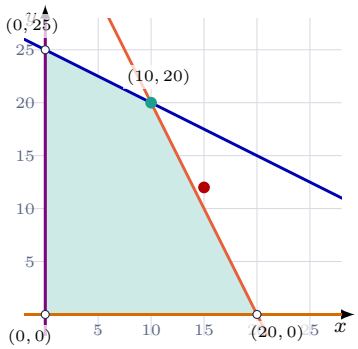
88. $20 \le 2(x+2x+1) \le 30 \Rightarrow 20 \le 6x+2 \le 30$: $\boxed{x \in [3, \frac{14}{3}]}$



(en cm)

89.

a) $2x+y \le 40$, $x+2y \le 50$, $x \ge 0$, $y \ge 0$. Vèrtexs $(0,0)$, $(20,0)$, $(10,20)$ i $(0,25)$.



b) $(15,12)$: $2 \cdot 15 + 12 = 42 > 40$: $\boxed{\text{no}}$ (falten hores). $(10,20)$: $40 \le 40$ i $50 \le 50$: $\boxed{\text{sí}}$, i aprofita totes les hores i tot el material.

90. $20t - 5t^2 > 15 \Rightarrow t^2 - 4t + 3 < 0 \Rightarrow t \in (1,3)$: $\boxed{\text{2 segons}}$

15 Ampliació i problemes

91. $P(-1) = 1+2+1 = \boxed{4}$

92. $P(x) = a(x-1)(x+2)(x-3)$; $P(0) = 6a = 12 \Rightarrow a = 2$: $\boxed{2x^3 - 4x^2 - 10x + 12}$

93.

a) $\dfrac{(x-3)(x^2+3x+9)}{(x-3)(x+3)} = \boxed{\dfrac{x^2+3x+9}{x+3}}$

b) $\dfrac{\frac{x-1}{x}}{\frac{(x-1)(x+1)}{x^2}} = \boxed{\dfrac{x}{x+1}}$

94. $2x^2 + 2(x+1)^2 = 5x(x+1) \Rightarrow x^2 + x - 2 = 0$: $\boxed{x = 1, x = -2}$ (cap anul·la els denominadors).

95. $x+3 = 1 + 2\sqrt{x-2} + x - 2 \Rightarrow \sqrt{x-2} = 2$: $\boxed{x = 6}$ ($3 - 2 = 1$ ✓).

96. $t = 2^x$: $2t^2 - 5t + 2 = 0 \Rightarrow t = 2, t = \frac{1}{2}$: $\boxed{x = 1, x = -1}$

97. $x^2 - 1 = 8 \Rightarrow x = \pm 3$; $x = -3$ no val: $\boxed{x = 3}$

98. $\left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 1 & -1 & 1 & 1 \\ 2 & 0 & a & 4 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & -2 & 0 & -2 \\ 0 & -2 & a-2 & -2 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & -2 & 0 & -2 \\ 0 & 0 & a-2 & 0 \end{array}\right)$

Si $a \neq 2$: $\boxed{\text{SCD}}$, $\boxed{(2,1,0)}$. Si $a = 2$: $\boxed{\text{SCI}}$, $\boxed{(2-\lambda, 1, \lambda)}$. (Mai no és incompatible.)

99. $x + y = 5$, $xy = 6 \Rightarrow x^2 - 5x + 6 = 0$: $\boxed{\text{2 cm} \times \text{3 cm}}$

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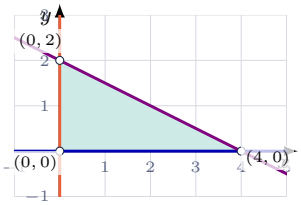
100. $x^2+1>0$ sempre, així que només cal $x^2-4\le 0$: $\boxed{x\in[-2,2]}$



101. $5000\cdot 1{,}04^t\ge 8000 \Rightarrow t\ge \dfrac{\log 1{,}6}{\log 1{,}04}\approx 11{,}98$: $\boxed{12 \text{ anys}}$

102. $p-2=3(f-2)$, $p+10=2(f+10)$: $\boxed{\text{pare } 38, \text{ fill } 14 \text{ anys}}$

103. $\boxed{x\ge 0,\ y\ge 0,\ x+2y\le 4}$ (la recta per $(4{,}0)$ i $(0{,}2)$ és $x+2y=4$, i O la compleix).



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